DOOR STATUS INDICATOR MODULE

KiCad Schematic, Footprint Selection and PCB Layout Report

Student Name	Himanshi Kaushik
Course / Branch / Section / Batch	B.Tech / ECE / B / 50
Institution	Krishna Institute Of Engineering and Technology
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1. Project Overview

The Door Status Indicator Module is a 5 V circuit intended to show whether a door is open or closed. A magnetic reed switch detects the position of a magnet mounted on the door. The circuit uses a green LED to indicate a closed door, a red LED to indicate an open door, and a buzzer to provide an audible warning when the door is open.

Design assumption

This report specifies a changeover (SPDT/Form-C) magnetic reed switch. Its common contact is connected to +5 V; the normally-open and normally-closed contacts select the closed-door and open-door indicator branches. A basic two-pin normally-open reed switch cannot directly provide both complementary outputs without adding an inverter or other logic stage.

2. Operating Principle

Door condition	Reed-switch state	Indicator	Buzzer
Closed (magnet near sensor)	COM → NO	Green LED ON	OFF
Open (magnet away from sensor)	COM → NC	Red LED ON	ON

Place the reed switch on the fixed door frame and the matching magnet on the moving door. The mechanical alignment and the switch's actuation direction must be checked against the selected part's datasheet.

3. Schematic Design

Use the following electrical connections when drawing the schematic in KiCad. Name the power nets +5V and GND. Use a connector for the 5 V input and a three-contact symbol for the SPDT reed switch.

Reference	Connection	Purpose
J1 – Power input	Pin 1: +5 V; Pin 2: GND	External regulated 5 V supply
SW1 – SPDT reed	COM: +5 V; NO: closed-door branch; NC: open-door branch	Selects the active indicator
D1 – Green LED	Selected branch → R1 → LED anode; LED cathode → GND	Door closed indication
D2 – Red LED	Selected branch → R2 → LED anode; LED cathode → GND	Door open indication
BZ1 – 5 V buzzer	Open-door branch → buzzer (+); buzzer (−) → GND	Audible open-door alert

Important implementation note: Confirm the reed switch contact convention and buzzer polarity from their datasheets before wiring. If the selected buzzer draws more current than the reed switch contact rating allows, use a suitable transistor driver stage and update the schematic and BOM. Do not connect an unknown buzzer directly until its rated voltage and current are checked.

4. Component Selection and Footprints

Choose footprints that match the exact physical parts being purchased. The examples below are common starting points; verify pad spacing, body dimensions, pin pitch and mounting requirements against each manufacturer's datasheet.

Component	Suggested KiCad footprint	Notes
J1 – 2-pin power connector	TerminalBlock or PinHeader 1x02, 2.54 mm	Choose screw terminal for a wired supply; check current rating.
SW1 – SPDT reed switch	Through-hole footprint matching the selected reed sensor	Often a custom footprint is needed; use datasheet dimensions.
D1, D2 – LEDs	LED_THT:LED_D3.0mm or LED_D5.0mm	Select 3 mm or 5 mm to match the physical LED.
BZ1 – 5 V buzzer	Buzzer footprint matching pin pitch and body diameter	Check polarity, pitch and mounting style.
R1, R2 – resistors	Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P7.62mm	Typical 1/4 W through-hole resistor; verify actual part.

5. Current-Limiting Resistor Calculation

For each LED, calculate its series resistor using: R = (VSUPPLY − VF) / ILED. For a 5 V supply, an assumed LED forward voltage of 2.0 V (red) or 2.1 V (green), and a target current of approximately 8–9 mA, a standard 330 Ω resistor is a reasonable starting value. Recalculate using the actual LED datasheet and desired brightness.

Example: red LED, R = (5 − 2.0) / 0.009 ≈ 333 Ω → choose 330 Ω. The green LED should be checked separately because its forward voltage may differ.

6. PCB Layout Guidelines

- Place the power connector near one board edge for easy wiring.
- Place the reed-switch connector or sensor at an edge where it can connect to the door-frame sensor.
- Keep the red and green LEDs near the board edge so their status is visible.
- Place the buzzer where its sound opening is not blocked by the enclosure.
- Keep traces short and use suitable trace widths for the buzzer current and power path.
- Use a continuous GND copper zone where practical, and add clear +5 V/GND labels.
- Observe the board manufacturer's clearance and track-width rules; run KiCad's DRC and resolve all errors.

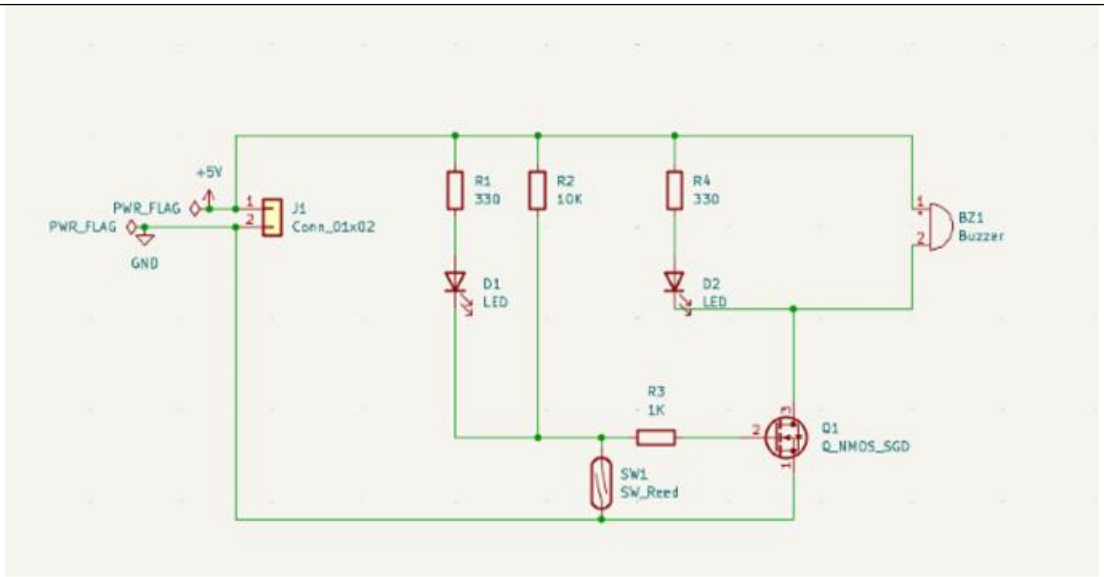
7. Bill of Materials (BOM)

Ref.	Qty.	Item	Value / specification
J1	1	Power connector	2-pin, 5 V input
SW1	1	Magnetic reed switch	SPDT / Form-C changeover
D1	1	Green LED	Through-hole; verify Vf
D2	1	Red LED	Through-hole; verify Vf
BZ1	1	Buzzer	Rated for 5 V DC; verify current
R1, R2	2	Current-limiting resistors	330 Ω, 1/4 W starting value

8. KiCad Design Evidence

Insert genuine screenshots from your completed KiCad project in the spaces below. The screenshots are intentionally left as placeholders because they must show your own schematic, assigned footprints and routed PCB.

8.1 Schematic Screenshot

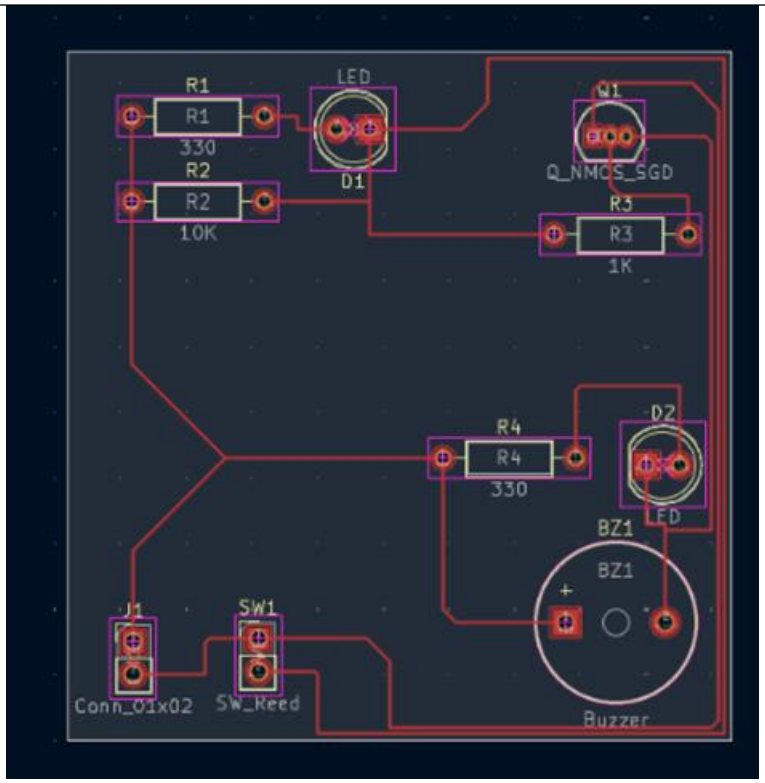


8.2 Footprint Assignment / Routed PCB Screenshot

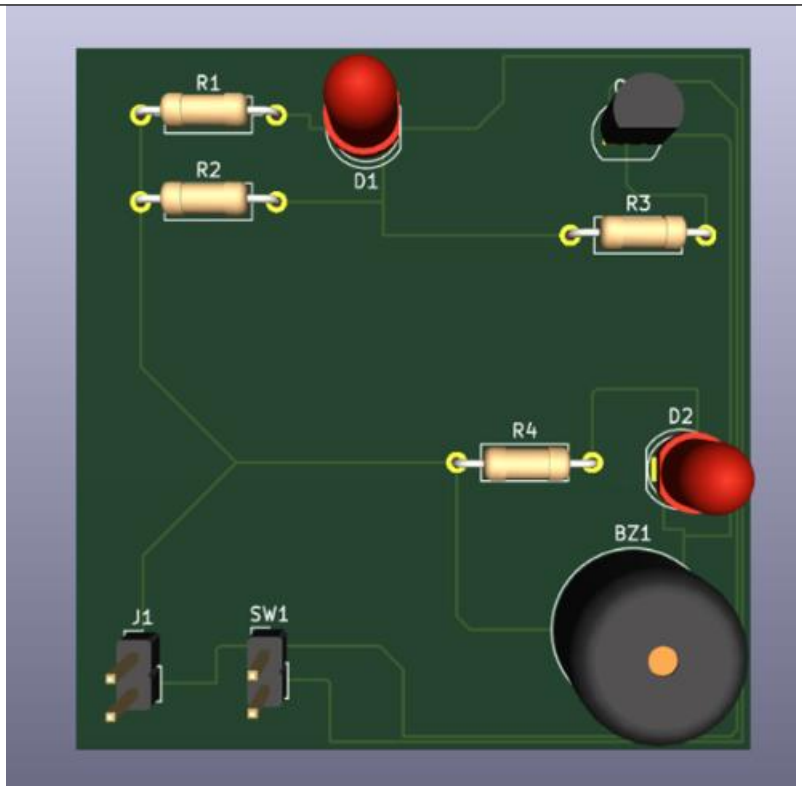
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 1      BZ1 -         Buzzer 5V : Buzzer_Beeper:Buzzer_12x9.5RM7.6
 2       D1 -         Green LED : LED_THT:LED_D5.0mm
 3       D2 -           Red LED : LED_THT:LED_D5.0mm
 4       J1 - Power Connector 5V : Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
 5       Q1 -           2N7000 : Package_TO_SOT_THT:TO-92_Inline
 6       R1 -              330 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
 7       R2 -              10K : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
 8       R3 -               1K : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
 9       R4 -              330 : Resistor_THT:R_Axial_DIN0207_L6.3mm_D2.5mm_P10.16mm_Horizontal
10      SW1 -      Reed Switch : Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical
```

9. 2D and 3D PCB Views

9.1 2D PCB View



9.2 3D PCB View



10. Verification and Final Submission Checklist

Before submission, complete the following checks:

☐ Confirm the reed switch is SPDT/Form-C and verify its contact action.

☐ Verify the buzzer voltage, current and polarity; add a driver stage if required.

☐ Check LED polarity and calculate each resistor from the selected LED datasheet.

☐ Assign footprints that match the actual components.

☐ Run Electrical Rules Checker (ERC) on the schematic.

☐ Run Design Rules Checker (DRC) on the PCB and resolve reported issues.

☐ Confirm there are no unconnected nets and that power/ground are correctly routed.

☐ Capture the schematic, footprint/routing, 2D and 3D screenshots.

☐ Export the BOM and include the final KiCad project files.

11. GitHub Project Link

GitHub Repository URL: https://github.com/kmhimansikaushik-dot/Door-Status-Indicator-Module

12. Conclusion

The proposed Door Status Indicator Module uses a magnetic reed switch to select a visual and audible indication for the door state. The design is intended for implementation as a 5 V PCB in KiCad. Final hardware release depends on confirming the exact reed-switch contact arrangement, LED characteristics, buzzer current and component footprints, followed by ERC, DRC and functional testing.